PREP 4, LESSONS 7–8

Power and wasted energy

Power as the rate of energy transfer; wasted energy; reducing it

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 State the law of conservation of energy. [1]

2 Describe how to find the spring constant from a graph of force (y-axis) against extension (x-axis). [1]

3 Name the store of energy associated with extending and compressing objects. [1]

Key ideas LESSONS 7–8

Show your working

4 A student climbs the stairs. They transfer 1500 J to their gravitational potential store in 6.0 s. Use $P = \frac{E}{t}$

Calculate the student's power. [3]

5 A 2000 W kettle is switched on for 120 s. Calculate the energy transferred. [2]

6 Where does most wasted energy end up? [1]

the kinetic store of the device	the thermal store of the surroundings	the chemical store of the fuel

7 Explain how oiling a bicycle chain reduces wasted energy. [1]

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